



National Supercomputing Mission, India

A first of its kind attempt to boost the country's computing power, the National Supercomputing Mission (NSM) is steered jointly by the Ministry of Electronics and IT (MeitY) and Department of Science and Technology (DST), and implemented by the Centre for Development of Advanced Computing (C-DAC), Pune and the Indian Institute of Science (IISc), Bengaluru. The target of the mission is set to establish a network of supercomputers ranging from a few teraflops (TFs) to hundreds of TFs and three systems with greater than or equal to three petaflops (PF) in academic and research institutions of national importance across the country by 2022. This network of supercomputers envisaging a total of fifteen to twenty PFs was approved in 2015 and was later revised to a total of 45 PFs (45000 TFs), a jump of six times more compute power within the same cost and capable of solving large and complex computational problems.

Objectives

- To make India one of the world leaders in supercomputing and to enhance India's capability in solving grand challenge problems of national and global relevance
 - To empower scientists and researchers with state-of-the-art supercomputing facilities and enable them to carry out cutting-edge research in their respective domains
 - To minimise redundancies and duplication of efforts, and optimise investments in supercomputing
 - To attain global competitiveness and ensure self-reliance in the strategic area of supercomputing technology

Current scenario

The first supercomputer assembled indigenously, called Param Shivay, was installed

in IIT (BHU) and was inaugurated by the Prime Minister. Similar systems like Param Shakti and Param Brahma were installed at IIT Kharagpur and IISER Pune. These are equipped with applications from domains like weather and climate, computational fluid dynamics, bioinformatics, and material science.

Eleven new systems are likely to be set up in different IITs, NITs, national labs, and IISERs across India by December this year, which will have many sub-systems manufactured and microprocessors designed in India, which will bring in a cumulative capacity of 10.4 PFs.

Spreading out the reach to the North-East region of the country, eight systems with a total compute power of sixteen PFs are being commissioned. Five indigenously designed systems with three PFs computing power will be installed at IIT Mumbai, IIT Chennai and Inter-University Accelerator Centre (IUAC) at Delhi with NKN as its backbone. It also includes an indigenously built twenty PFs system at C-DAC Pune, and a hundred PFs artificial intelligence (AI) supercomputing system. One midlevel 650 TFs system is also to be installed at C-DAC Bengaluru to provide consultancy to startups, SSIs and MSMEs.

Geared to provide supercomputing facility to about sixty to seventy institutions nationwide and more than thousands of active researchers, academicians, and so on, NSM has gathered momentum and is moving fast not only towards creating a computer infrastructure for the country but also to build the capacity of the country to develop the next generation of supercomputer experts.

Application areas

Climate Modelling, Weather Prediction, Aerospace Engineering including CFD, CSM, CEM, Computational Biology, Molecular Dynamics, Atomic Energy Simulations, National Security/Defence Applications, Seismic Analysis, Disaster Simulations and Management, Computational Chemistry, Computational Material Science and Nanomaterials, Discoveries beyond Earth (Astrophysics), Large Complex Systems Simulations and Cyber-Physical Systems, Big Data Analytics, Finance, Information Repositories/Government Information Systems **EFY**

Source: <https://dst.gov.in/>, <https://nsmindia.in>